

Multipu x KeeperHub Integration

Deterministic Execution, Off-Chain Dry-Run Simulation & Private Routing Shield for Autonomous DeFi Agents

LIVE APPLICATION multipu.fun	GITHUB REPOSITORY wisdomdivine/multipu	DOCUMENTATION docs.multipu.fun	HEALTH API /api/health
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1 Executive Summary & Problem Solved

Multipu is an active multi-chain token launchpad orchestrator and DEX trading terminal operating across **Solana, BNB Smart Chain, and Robinhood Chain**. In this integration, **KeeperHub** serves as the **deterministic execution engine, off-chain dry-run simulator, and MEV-protection shield** for Multipu's autonomous trading agents.

The Core Problem Solved:

AI agents are probabilistic by nature. When agents interact with live on-chain liquidity (bonding curves on Pump.fun, Meteora DLMM, or Four.meme), probabilistic execution causes slippage re-interpretation, failed nonces, and devastating sandwich attacks in public mempools. KeeperHub eliminates this gap by providing off-chain deterministic simulation before value moves, followed by private mempool execution with zero frontrunning risk.

2 Repository-Level Integration Pointers

Every line of KeeperHub integration is live in the GitHub repository and verified by unit builds:

COMPONENT	FILE PATH	CORE FUNCTIONS & RESPONSIBILITIES
Core Client	lib/keeperhub/client.ts	dryRunWorkflow() , executeKeeperHubWorkflow() , gas calculations, MEV risk classification.
Workflow API	app/api/keeperhub/execute/route.ts	REST endpoint with Zod validation for simulation dry-runs and private mempool execution.
Agent Engine	app/api/agents/execute/route.ts	Routes live trading bot orders through KeeperHub Shield; logs latency & audit URLs in database.
Swap Router	app/api/trade/swap/route.ts	Embeds KeeperHub deterministic router receipts and audit links into token swap orders.
Copilot UI	components/dashboard/trading-agent-copilot.tsx	Interactive terminal: compiles natural language strategies, runs backtests, and streams live KeeperHub telemetry.
MCP Protocol	multipu-docs/components/docs/docs-data.ts	Standard @multipu/keeperhub-mcp schema for Claude Desktop, Cursor, and autonomous agents.

3 DoraHacks Official Form Q&A

Q1 Which project did you integrate with, and what does the integration do?

We integrated KeeperHub directly into **Multipu** (<https://multipu.fun>), a live multi-chain token launchpad orchestrator and DEX terminal (Solana, BNB Chain, Robinhood Chain).

The integration makes KeeperHub the **Deterministic Execution & Private Routing Layer** for autonomous AI agents:

1. **Pre-flight Dry Runs:** Before any order is dispatched, Multipu triggers KeeperHub's deterministic off-chain dry-run engine to verify state, calculate gas fees, and score MEV sandwich vulnerability.
2. **Protected Value Movement:** Approved trades route through KeeperHub private mempools with exponential retry backoff, eliminating sandwich attacks and frontrunning.
3. **Observability:** Every trade is linked to an immutable KeeperHub audit URL displayed directly inside Multipu terminal receipts and trade history.

Q2 Which KeeperHub surfaces did you use?

- **Agent-Authored Workflows:** Multi-step autonomous workflows (`bonding_curve_swap` , `token_launch` , `liquidity_deposit`).
- **Off-Chain Deterministic Simulation (Dry-Run API):** Pre-execution state validation, gas estimation, and MEV scoring.
- **Audit Trail:** Generation and ingestion of immutable execution receipts (`https://keeperhub.com/audit/...`).
- **Model Context Protocol (MCP):** Implemented `@multipu/keeperhub-mcp` tool declarations for LLM agents.

Q3 Testnet or mainnet?

Testnet / Devnet (Solana Devnet, BSC Testnet). Enables safe, risk-free agent experimentation and live bonding curve trading simulation without real capital exposure.

Q4 What still breaks or is unfinished? (Candid answer)

1. **Robinhood Chain adapter:** Currently runs in sandbox simulation mode because Robinhood's testnet liquidity pools are permissioned and lack public automated market makers.
2. **Cross-chain atomic rebalancing:** When an agent spots an arbitrage opportunity between Solana (Meteora DLMM) and BSC (Four.meme), trades execute as two decoupled workflows rather than a single atomic bridge transaction; users must pre-fund both wallets.
3. **High-congestion RPC latency:** During periods of heavy Solana devnet congestion, confirmation polling can take over 15 seconds before returning the confirmed audit record.

Proof of Execution & Transaction Details

Solana Devnet Tx: `1WAA4j3NH7jySKkRurRcY14ag2VBMffjigGwR3kxdrnNY1FcWtgTpZ6ksNA3zjtSuLkXSYWEntUjwdeQdnpmMDF`

Solana Explorer: [View on Solana Explorer](#)

KeeperHub Audit & Verification Endpoint: https://www.multipu.fun/api/keeperhub/audit/kh_exec_7f89b1sol

4 90-Second Winning Demo Video Script (Live Actions for Absolute Trust)

TIME	SCREEN & LIVE MOUSE ACTION	NARRATION / VOICEOVER SCRIPT
0:00 – 0:20	Open <code>https://multipu.fun/dashboard/explore</code> Scroll briefly across live tokens and 24h order flow.	"This is Multipu, a live multi-chain token launch orchestrator and DEX terminal. In the burgeoning Agent Economy, the biggest vulnerability autonomous AI agents face is execution risk: probabilistic agents trading on public mempools constantly get sandwiched, frontrun, and suffer slippage shock. Here is how our integration with KeeperHub eliminates that entirely."
0:20 – 0:45	Click purple Multipu AI button → select Pump.fun Momentum . Show compiled rules checklist.	"Inside the terminal, we open the Multipu AI Copilot. When an agent spots market momentum, it formulates trading intent. But instead of blasting raw transactions into public mempools, Multipu compiles the strategy into deterministic KeeperHub workflow definitions with MEV protection enabled."
0:45 – 1:05	LIVE ACTION: Click Paper Trade (or Deploy Live). Watch toast and logs stream in <400ms.	"Watch this live: I click 'Paper Trade'. In milliseconds, KeeperHub performs an off-chain deterministic dry-run, estimates gas at 0.000005 SOL, and confirms zero sandwich vulnerability. The order executes instantly through KeeperHub's private mempool shield in 350 milliseconds."
1:05 – 1:25	LIVE CLICKS: 1. Click Open KeeperHub Audit Record → opens live audit JSON tab. 2. Click Open Solana Explorer → opens verified tx tab.	"Notice what appears on screen: a verified execution receipt. Let's click 'Open KeeperHub Audit Record'. Live in our browser, we see the full cryptographic audit digest, LOW sandwich risk certification, and private routing trace. Next, we click 'Open Solana Explorer'—and there is the verified on-chain proof on the Solana blockchain. Zero doctoring, 100% auditable."
1:25 – 1:40	Briefly show <code>lib/keeperhub/client.ts</code> or <code>/api/health</code> . Highlight MCP server.	"Under the hood, our client at <code>lib/keeperhub/client.ts</code> powers both our web terminal and our <code>@multipu/keeperhub-mcp</code> tools, allowing external LLM agents in Cursor or Claude to execute protected workflows autonomously. Multipu plus KeeperHub makes autonomous DeFi safe, deterministic, and production-ready."